

The moderation of ambivalence on attitude–intention relations as mediated by attitude importance

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Abstract

Attitudinal ambivalence has been found to moderate attitude–intention relations. However, no prior work has investigated the mechanisms by which this moderation effect occurs. The present research attempted to address this issue. Across two studies, there was evidence that an Ambivalence $\times$ Attitude interaction was mediated through judgements about attitude importance. Additionally, the present research ruled out the possibility that attitude certainty, a factor that is often found to be positively related to attitude importance, was not responsible for the observed mediating effects of this latter variable. While replicating previous evidence supporting the moderating properties of ambivalence on attitude–intention relations, the current research sheds light on the critical role that attitude importance plays in this relationship. Copyright © 2006 John Wiley & Sons, Ltd.

The theory of planned behaviour (Ajzen, 1985, 1991) is one of the dominant accounts of the relationship between attitude and behaviour in social psychology. According to the model, the most proximal determinant of an individual's behaviour is the intention to perform it. Social psychological theory and research has long had an interest in the factors that might influence the strength of the relationship between attitude to behavioural intention (Eagly & Chaiken, 1993; Kraus, 1995; Petty & Krosnick, 1995; Wicker, 1969). From this prior work, a key prediction is that stronger attitudes should be more likely to predict intention than weaker attitudes (Converse, 1995).

Other attitude theorists have suggested that the simultaneous presence of positive and negative evaluations of the same attitude object (Kaplan, 1972; Thompson, Zanna, & Griffin, 1995), namely, attitudinal ambivalence (Scott, 1966, 1969), may in fact *moderate* the strength of attitude–intention relations (e.g. Eagly & Chaiken, 1993; Thompson et al., 1995). Specifically, attitudes have been proposed to be more predictive of intentions under low levels of ambivalence (see Conner & Sparks, 2002; Jonas, Broemer, & Diehl, 2000, for discussions), a hypothesis empirically supported by a number of studies (e.g., Armitage & Conner, 2000; Conner et al., 2002; Moore, 1973, 1980; Sparks, Conner, James, Shepherd, & Povey, 2001; Sparks, Harris, & Lockwood, 2004; Sparks, Hedderley, & Shepherd, 1992; see Conner & Sparks, 2002; Jonas et al., 2000, for a review and discussion).

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As a moderator, ambivalence may reflect processes that explain its influence on the issue of how attitudes come to be translated into intentions (Bagozzi, 1992). Therefore, the present study was designed to investigate this issue. Relevant to the present research is previous work showing that attitude strength does not involve a single underlying dimension (e.g. Krosnick, Boninger, Chuang, Berent, & Carnot, 1993; Prislin, 1996). Specifically, strength properties of the attitudinal structure (e.g. accessibility or extremity) have been shown to be distinct from the so-called meta-attitudinal strength properties, which reflect subjective impressions about one's attitude (Bassili, 1996; see Krosnick & Petty, 1995, for a review). Of particular relevance for the current work are the meta-attitudinal strength properties of (a) experienced ambivalence, a measure tapping people's awareness of their own accessible attitudinal inconsistencies toward a given attitude object (Jonas et al., 2000), and (b) attitude importance, or the subjective sense of concern about an attitude and the psychological significance that one attaches to it (Boninger, Krosnick, & Berent, 1995).

Unlike other attitude strength properties, previous research (e.g. Visser, Krosnick, & Simmons, 2003) has shown that attitude importance may be characterised as a *motivational* variable propelling the individual to express one's attitude by acting in accordance with it (Craig, Martinez, Kane, & Gainous, 2002; Krosnick & Petty, 1995). In the current research, therefore, we examined the role of attitude importance as a proximal antecedent of intention rather than focusing on other attitude strength properties. Specifically, we proposed that attitude importance as a proximal antecedent of intention may be mediating the moderating effect of the more distal predictor of attitudinal ambivalence (*mediated moderation*: Baron & Kenny, 1986; Wegener & Fabrigar, 2000).

Clearly, we are aware that our mediated-moderation hypothesis should be based on firm theoretical and/or empirical grounds for predicting, in the first place, that ambivalence influences attitude importance (cf. Baron & Kenny, 1986; Wegener & Fabrigar, 2000). Indeed, attitudinal ambivalence has been proposed to be an antecedent of attitude importance (e.g. Armitage, 2003). Unfortunately, no direct empirical findings are available in this respect. However, in general, attitude strength properties of the conceptually same (structural or meta-attitudinal) type have often been found to be inter-related (e.g. Bassili, 1996; Eagly & Chaiken, 1995; Jonas et al., 2000; Krosnick et al., 1993; Pomerantz, Chaiken, & Tordesillas, 1995). For example, previous research has found stronger ambivalence to be associated with lower attitude certainty (e.g. Jonas, Diehl, & Broemer, 1997), an attitude strength property belonging to the same (meta-attitudinal) type as attitude importance (Bassili, 1996; Jonas et al., 2000). This notion has also been supported by a number of studies. For instance, Lavine, Borgida, and Sullivan (2000, Study 1 and 3) found support for the negative relationship we assume between ambivalence and importance ($r_s = -0.28$ and -0.38 , respectively).

The rationale for our mediated moderation hypothesis is based on recent work by McGregor, Newby-Clark, and Zanna (1999). These researchers have characterised ambivalence as a form of internal attitudinal inconsistency that has negative psychological consequences that are similar to those of cognitive dissonance (Festinger, 1957). Accordingly, they have argued that people will be motivated to reduce ambivalence to eliminate the unpleasant state elicited by their own ambivalence (Newby-Clark, McGregor, & Zanna, 2002). In the terminology of cognitive dissonance theory, we argue that ambivalent attitude-holders face a typical 'forced choice' situation: they cannot change either the internally inconsistent nature, or the degree, of their ambivalence towards the object. As a consequence, we propose that one of those strategies people will be motivated to use to reduce the unpleasant state elicited by their own ambivalence is to discount the importance of their ambivalent attitude object by perceiving it as less self-relevant. The less important the object of the ambivalent attitude, the weaker the resulting dissonance and the unpleasant state associated with it. In turn, since people are not likely to express one's attitude by acting in accordance with it to the extent that they perceive low attitude importance (e.g. Craig et al., 2002; Krosnick & Petty, 1995; Visser et al., 2003),

the latter should mediate the effects of the moderating role of attitudinal ambivalence on behavioural intentions.

Two studies were conducted to test this mediated moderation hypothesis. In both studies, the two particular target behaviours (i.e. waste recycling and eating a low-fat diet) were selected because they are behavioural domains where attitudinal ambivalence is prevalent (Costarelli & Colloca, 2004; Sparks et al., 1992). Accordingly, we anticipated that each of these behaviours would provoke substantial levels of attitudinal ambivalence at least for a significant portion of our samples.

STUDY 1

Method

Participants and Procedure

Sears (1986, 1987) suggested that over-reliance on convenience samples of university students may have lead social psychology to paint a misleading portrait of human nature. Because we aimed at generalising our results beyond such samples, we set out to test our predictions in a sample that was as heterogeneous as possible.

To this end, data were collected by telephone by a staff of trained interviewers. Telephone numbers were generated by attaching randomly generated five digits to the end of the prefix for Trento-area telephone numbers (Waksberg, 1978). The overall response rate was 88.3% out of the eligible persons contacted. The resulting research sample was composed of 84 female and 26 male adult residents of the Trento, Italy, metropolitan area, mean age 50.2 ($SD = 19.1$; range 18–94 years). There were no significant effects for participant age or sexual gender. Therefore, we will not discuss these variables further.

The target behaviour was ‘waste recycling’. Participants were asked to answer questions about waste recycling which were presented in a fixed order.

Measures

Because of the limitations of telephone interviewing, we used rating scales with no more than four points.

First, *attitude* was assessed by asking participants to think about waste recycling, providing ratings (1 = very cold, 2 = cold, 3 = warm, 4 = very warm) indicating how cold or warm they felt, a measure that has been successfully used in past research (‘thermometer rating’: e.g. Campbell, 1971; Miller, Miller, & Schneider, 1980).

Next, *ambivalence* was measured by unipolar split-semantic differential scales (Kaplan, 1972). The items assessed positive and negative evaluations separately. The form of these questions was taken from Thompson and colleagues (1995) and included ‘Considering only the unfavourable qualities of waste recycling and ignoring the favourable characteristics, how unfavourable is your evaluation of waste recycling?’ (‘*Not at all unfavourable-extremely unfavourable*’), and ‘Considering only the favourable qualities of waste recycling and ignoring the unfavourable characteristics, how favourable is your evaluation of waste recycling?’ (‘*Not at all favourable-extremely favourable*’). To calculate ambivalence towards the target behaviour (waste recycling), these two measures were inserted into the most widely used ambivalence formula, namely, the ‘Griffin’ method (see Thompson et al., 1995).

According to this method, ambivalence is calculated as the polarisation of the positive (P) and negative (N) judgements, minus the absolute difference between the two, or $(P + N)/2 - |P - N|$.

Then, *attitude importance* and *attitude certainty* were assessed using the items 'How important is the issue of waste recycling to you personally?' and 'How certain are you of the attitude you hold on the issue of waste recycling?' respectively (1 = *not at all*, 4 = *extremely*), as adapted from Visser and colleagues (2002).

Behavioural intention was subsequently assessed using the item: 'I intend to recycle my waste in the next week' (1 = *not at all*, 4 = *definitely*).

Finally, participants were asked to indicate their age and gender, were debriefed and thanked.

Results and Discussion

We expected that attitudinal ambivalence would moderate the influence of attitude on participants' behavioural intention through the mediating effect of attitude importance. According to Baron and Kenny (1986), mediation can be said to occur when three conditions are satisfied: first, the mediated (independent) variable should be associated with the dependent variable; second, the mediated (independent) variable should be associated with the mediating variable; and third, in a regression of the dependent variable on the both the mediated variable and the mediator, the effect of the mediated variable should be reduced, whereas the effect of the mediator should be statistically significant. To test such mediated-moderation prediction, following Baron and Kenny (1986) and Wegener and Fabrigar (2000), we conducted three regressions.

Specifically, first, we tested the moderating effect of ambivalence. To this end, we employed moderated (hierarchical) regression analysis. Preliminary correlational analyses showed that attitude and ambivalence did not correlate with each other, $r = -0.15$, $p > 0.12$. This finding ruled out the possibility that predictor (attitude) and moderator (ambivalence) were confounded. Subsequently, we computed mean-centred scores for these variables to minimise problems of multicollinearity (Aiken & West, 1991). A moderator effect is evident if the interaction between moderator variable and moderated variable is reliable while the independent effect of each are statistically controlled (Baron & Kenny, 1986). Table 1.

Concerning main effects, the moderated regression analysis revealed that more favourable attitudes were associated with stronger behavioural intentions, $\beta = 0.47$, $p < 0.001$. Additionally, increases in ambivalence were found to be associated with decreases in intention, $\beta = -0.24$, $p < 0.01$. The final equation was statistically significant ($F = 13.32$, $p < 0.001$), with the two predictors accounting for 19% of the adjusted variance in intentions. Most importantly, the attitude by ambivalence interaction effect on intention was also significant ($p < 0.05$) and its addition to the equation resulted in a statistically significant improvement in prediction over and above the effects of the other terms, $R^2_{\text{change}} = 0.04$, $F_{\text{change}} = 4.52$, $p < 0.05$. The direction of the regression coefficient for the interaction term indicated that, as ambivalence increased, the strength of the attitude-intention relation weakened, $\beta = -0.28$. In terms of the mediated moderation hypothesis, these results satisfy the first criterion for mediation because the interaction between ambivalence and attitude was significantly associated with behavioural intention.

Mediated-Moderation Analysis

Next, we tested the second criterion for mediation by regressing attitude importance (the proposed mediating variable) on attitude, ambivalence, and attitude by ambivalence interaction term. The results of the analysis indicated that more favourable attitudes were associated with greater attitude

Table 1. Moderating effects of attitudinal ambivalence on the attitude–intention relation as mediated by attitude importance

	First step			Second step		
	R^2_{adj}	F	t	R^2_{adj}	F	t
1. Predicting intention						
Attitude	0.19	13.32***	3.84***	0.23	17.84***	3.32**
Ambivalence			−2.73*			−2.20*
Attitude × Ambivalence						−2.53*
2. Predicting importance						
Attitude	0.10	6.52**	2.48*	0.17	9.83***	2.30*
Ambivalence			−2.16*			−1.90
Attitude × Ambivalence						−2.04*
Only one step						
	R^2_{adj}	F		t		
3. Predicting intention						
Attitude	0.48	25.42***		2.28*		
Ambivalence				0.96		
Importance				2.30*		
Attitude × Ambivalence				−1.57		

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

importance, $\beta = 0.24$, $p < 0.05$. Additionally, increases in ambivalence were associated with decreases in attitude importance, $\beta = -0.20$, $p < 0.05$. The final equation was statistically significant ($F = 6.52$, $p < 0.01$), with the two predictors accounting for 10% of the adjusted variance in attitude importance. Most importantly, the attitude by ambivalence interaction on attitude importance was also significant ($p < 0.05$), and its addition to the equation resulted in a statistically significant improvement in prediction over and above the effects of the other terms, $R^2_{\text{change}} = 0.07$, $F_{\text{change}} = 3.31$, $p < 0.05$. The direction of the regression coefficient for the interaction term indicated that, as ambivalence increased, the strength of the attitude–importance relation became weaker, $\beta = -0.42$. These findings satisfy Baron and Kenny's second criterion, namely that the mediated independent variable (in this case, the attitude by ambivalence interaction term) was associated with the mediator (attitude importance).

Finally, to test the third criterion for evidence of mediation, behavioural intentions were regressed on attitude, ambivalence, attitude importance, and the interaction term between attitude and ambivalence.

This analysis revealed that increases in intentions were associated with increases in attitude importance ($\beta = 0.38$, $p < 0.05$) and attitude ($\beta = 0.34$, $p < 0.05$), whereas ambivalence was a non-significant predictor of intention, $\beta = 0.08$, $p > 0.34$. Most importantly, the attitude by ambivalence interaction effect on intention was not significant, $\beta = -0.17$, $p > 0.12$. The final equation was statistically significant ($F = 25.42$, $p < 0.001$), with the predictors accounting for 48% of the adjusted variance in intention. A subsequent Sobel test (Sobel, 1982) indicated that the reduction in the effect of the attitude by ambivalence interaction term was significant, $z = -2.01$, $p < 0.05$.

These results provide clear, albeit preliminary, support for our mediated–moderation hypothesis that ambivalence influences the strength of the attitude–intention relationship via attitude importance. To test for the potential role of attitude certainty as a mediator of the effects of attitude importance, equivalent analyses were subsequently performed including attitude certainty in the model as a predictor of intention. These analyses revealed that certainty was a significant predictor of intentions ($\beta = 0.23$, $p < 0.01$), such that the more certain people were about their attitudes, the stronger their

intentions. Most importantly, the previously identified effect of attitude importance on intentions remained significant ($\beta = .47$, $t = 6.03$, $p < 0.001$) even when controlling for the effects of attitude certainty. The implication of these findings is that the effects of attitude importance are independent of the effects of attitude certainty.

One limitation of Study 1 is that it examined only one specific target behaviour. Further, our exclusive use of one of the two existing types of ambivalence measures might have biased the test of our hypotheses. Study 2 was designed to test for this possibility using a different target behaviour and a different method for assessing ambivalence to the ones employed in Study 1.

STUDY 2

In Study 2, we set out to test whether we could replicate the pattern of results found in Study 1 in a different target behavioural domain (i.e. eating a low-fat diet) and using another type of ambivalence measure, namely, experienced ambivalence. This latter way of assessing ambivalence (see below) reflects participants' conscious acknowledgement that they hold a mixed evaluation (i.e. both positive and negative) of the attitude object. Experienced ambivalence is typically only moderately related to the type of ambivalence measures used in Study 1 (i.e. formula-based ones; see Priester & Petty, 1996; Thompson et al., 1995, for a review). Thus, use of both types of ambivalence measures ensures a complete assessment of the impact of attitudinal ambivalence on psychological phenomena (cf. Conner & Sparks, 2002; Jonas et al., 2000).

Method

Participants and Design

In Study 2, 27 male and 58 female volunteers from Trento, Italy, whose ages ranged from 16 to 70 years ($M = 38.8$, $SD = 10.7$) were requested to complete a questionnaire. There were no significant effects for participant age or sexual gender. Therefore, we do not discuss these variables further.

The design was also identical to Study 1, with one difference. Specifically, the target behaviour was 'eating a low-fat diet', a behaviour in which many people engage and where conflicting motivations are prevalent. Given the null findings in Study 1 for the impact of attitude certainty as potential alternative mediator of the moderating effect of ambivalence on the attitude–intention relationship, in Study 2 attitude certainty was not assessed.

Procedure and Measures

Compared to the previous study, the procedure was the same, with the exception that the ambivalence measure was different to the one used in Study 1 and additional items were employed to assess the other constructs. Additionally, to enhance measurement validity, responses were not expressed on 4-point Likert scales as in Study 1 but on 6-point ones. Specifically, in Study 2, we assessed experienced rather than formula-based ambivalence. Experienced ambivalence was measured by using three items (*Cronbach's* $\alpha = 0.75$): 'Concerning eating a low-fat diet, I'm both for and against', 'On eating a low-fat diet, I feel myself conflicted', and 'I have mixed emotions (both positive and negative) on the issue of eating a low-fat diet, all at the same time'.

Attitude was measured using two semantic differential scales with no neutral point: negative–positive, and unfavourable–favourable ($\alpha = 0.92$; ‘My evaluation of the issue of eating a low-fat diet is $-3 = \text{extremely negative/unfavourable}$ to $+3 = \text{extremely positive/favourable}$).

Attitude importance was assessed using three items ($\alpha = 0.77$): ‘How important is the issue of eating a low-fat diet to you personally?’ ‘How much do you care about the issue of eating a low-fat diet’, and ‘How much does the issue of eating a low-fat diet matter to you personally?’ (1 = *not at all*, 6 = *extremely*).

Behavioural intention was subsequently assessed using three items ($\alpha = 0.94$): ‘I intend/plan/want to eat a low-fat diet in the next week’ (1 = *not at all*, 6 = *extremely*), as adapted from Armitage (2003) for the current behaviour.

Scale scores were again computed by averaging responses across the respective items.

Results and Discussion

The same statistical analyses were performed as in the previous study. Again, we expected that attitudinal ambivalence would moderate the influence of attitude on participants’ behavioural intention through the mediating effect of attitude importance. To test such a mediated–moderation prediction, we again conducted three regression analyses.

Preliminary correlational analyses showed that attitude and ambivalence did not correlate with each other, $r = 0.01$, $p > 0.90$. This finding ruled out the possibility that predictor (attitude) and moderator (ambivalence) were confounded. Subsequently, we computed mean-centred scores for these variables to minimise problems of multicollinearity (Aiken & West, 1991). Again, in terms of the predicted moderating effect of ambivalence on attitude–intention relations, intention was regressed on attitude (Step 1), ambivalence (Step 1), and their interaction term (Step 2). Table 2.

Concerning main effects, results for revealed only an effect of attitude such that more favourable attitudes were associated with stronger behavioural intentions, $\beta = 0.47$, $p < 0.001$. Most importantly, again, the attitude by ambivalence interaction effect on intention was also significant, $\beta = -0.29$, $p < 0.01$. Addition of the interaction term to the model yielded a significant contribution to the regression equation, $R^2_{\text{adjusted}} = 0.36$; $F(2, 81) = 16.92$, $p < 0.001$, $R^2_{\text{change}} = 0.07$, $F_{\text{change}} = 9.85$, $p < 0.01$. This finding indicated that ambivalence was a moderator of the attitude–intention relationship. To interpret the interaction, we examined the regression lines at two levels of the hypothesised moderator (ambivalence): lower ambivalence (one standard deviation below the mean) and higher ambivalence (one standard deviation above the mean).

These analyses revealed that when ambivalence was lower, attitude positively predicted behavioural intentions, $\beta = 0.78$, $p < 0.0001$. In contrast, in the higher-ambivalence group, attitude did not significantly predict intentions, $\beta = -0.18$, $p > 1$. These findings show that attitudes are more predictive of behavioural intentions to the extent that ambivalence is lower. Moreover, these results satisfy the first criterion for mediation because the interaction between ambivalence and attitude was significantly associated with behavioural intention.

Next, we tested the second criterion for mediation by regressing attitude importance (the proposed mediating variable) on attitude, ambivalence, and their interaction term. The results indicated only a main effect of attitude ($p < 0.001$), such that attitude importance was higher to the extent that the attitude was more favourable, $\beta = 0.57$. Most importantly, the expected attitude by ambivalence interaction on attitude importance was also found, $\beta = -0.18$, $p < 0.05$. Addition of the interaction term to the model yielded a significant contribution to the regression equation, $R^2_{\text{adjusted}} = 0.39$, $F = 18.85$, $p < 0.001$; $R^2_{\text{change}} = 0.03$, $F_{\text{change}} = 4.06$, $p < 0.05$.

Table 2. Moderating effects of attitudinal ambivalence on the attitude-intention relation as mediated by attitude importance

	First step			Second step		
	R^2_{adj}	F	t	R^2_{adj}	F	t
1. Predicting intention						
Attitude	0.29	7.07***	5.09***	0.36	16.92***	4.05***
Ambivalence			-1.18			-1.12
Attitude $\times$ Ambivalence						-3.14**
2. Predicting importance						
Attitude	0.36	14.79***	6.25***	0.39	18.85***	4.52***
Ambivalence			-1.88			-1.04
Attitude $\times$ Ambivalence						-2.01*
Only one step						
	R^2_{adj}	F		t		
3. Predicting intention						
Attitude	0.61	33.93***		1.23		
Ambivalence				-1.32		
Importance				7.23***		
Attitude $\times$ Ambivalence				-2.34*		

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Finally, to test the third criterion, behavioural intentions were regressed on attitude, ambivalence, attitude importance, and the interaction term between attitude and ambivalence. This analysis revealed that attitude importance was a significant predictor of intention ($\beta = 0.64$, $p < 0.001$), but the main effects of ambivalence ($\beta = -0.01$, $p > 0.1$) and attitude ($\beta = 0.11$, $p > 0.1$) were non-significant. The attitude by ambivalence interaction effect on intention was also statistically significant, $\beta = -0.17$, $p < 0.05$. The final equation was statistically significant ($F = 33.93$, $p < 0.001$), with the predictors accounting for 61% of the adjusted variance in intention. A subsequent Sobel test (Sobel, 1982) indicated that the reduction in the extent of the attitude by ambivalence interaction was significant, $z = -2.05$, $p < 0.05$.

In sum, once again, attitude importance mediated the moderating effect of ambivalence on the strength of the attitude–intention relation. More important, these effects were demonstrated in a different research sample and even using a different ambivalence measure and attitude object (behaviour). This increases the generalisability of the present findings.

GENERAL DISCUSSION

Previous research on attitude–intention relations has been concerned predominantly with variables affecting the strength of the relationship between attitudes and intentions. A growing theoretical and empirical literature has focused on the role of attitudinal ambivalence as one such moderator. However, no prior work has investigated the mechanisms by which this moderation effect occurs. The present research attempted to address this gap of knowledge by focusing on the concept of attitude importance as a potential mediating variable.

In line with predictions, across the two studies presented, consistent mediated–moderation effects were shown for two distinct attitude objects and ambivalence measures. Additionally, the present research ruled out the possibility that a factor that has been found to be positively related to attitude importance, namely, attitude certainty was responsible for the predicted mediating effects of attitude importance.

This work contributes to this literature in two ways. First, the current findings replicate previous evidence supporting the moderating properties of ambivalence on attitude–intention relations. More important, this work expands the scope of such research by identifying the critical role that attitude importance plays in this respect. Specifically, the present research provides novel evidence that ambivalence acts as a moderator of the predictive validity of attitudes through its relationship with attitude importance, which is the mediator of the moderating effect of ambivalence.

These results are consistent with previous meta-analytic evidence provided by Cooke and Sheeran (2004) showing that attitude importance ('involvement') is a stronger moderator of attitude–intention relations than ambivalence ($r_s = 0.17$ and 0.06 , respectively). More specifically, higher attitude importance participants showed greater consistency between attitude and intention than lower attitude importance participants ($r_s = 0.57$ and 0.31 , respectively). Additionally, our findings are in line with previous suggestions that ambivalence is an antecedent of attitude importance (e.g. Armitage, 2003).

The evidence provided in this research supports the view that ambivalence is an important variable for those involved in attitude research to consider (cf. Breckler, 1994; Cacioppo, Gardner, & Berntson, 1997; Conner & Sparks, 2002; Costarelli & Palmonari, 2003; Jonas et al., 2000; Priester & Petty, 1996; Thompson et al., 1995). We would anticipate that the present findings are potentially relevant to a number of health protecting behaviours, which might be characterised by considerable levels of ambivalence (e.g. safer sex practices; exercise).

There were some limitations to this research that need acknowledgement. Although the replication of our findings across two studies is a strength of the current research, a complete replication of this work with multi-item scales would be desirable. Additionally, future research should establish the generality of the effects reported in the current investigation across a range of different behavioural intentions by conducting within-participants studies. Moreover, to strengthen support for our proposed mediated moderation hypothesis, future research should seek to experimentally manipulate attitudinal ambivalence. Finally, the extent to which similar effects are found with measures of ambivalence based on cognitive or affective within-dimension conflicts or cognitive-affective cross-dimension conflicts (Lavine, Thomsen, Zanna, & Borgida, 1998; MacDonald & Zanna, 1998) is worthy of study.

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